

Extended Abstract

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Motivation The burgeoning domain of Human-Centered AI is cautiously optimistic about the potential of the field of collaborative robotics. Spurred by recent advances in robotics and machine learning technology, robots are increasingly present in domestic and industrial spaces, which raises concerns about how we can build autonomous systems that are "cooperative and well-aligned [with] humans' needs, goals, and values" Littman et al. (2022). Simulation offers a dynamic benchmark environment in which open-ended human-sourced tasks can drive the development and evaluation of human-centric collaborative robots Li et al. (2024). However, the extent to which human-described tasks can be reconstructed for benchmarks face severe limitations: the simulation environment lacks requisite annotations to provide concrete evaluations of robot performance on open-ended task definitions. To this end, we explore a means of learning semantic scene representations which can be leveraged for open-ended environment annotation for the evaluation of generalist robot learning on arbitrary tasks.

Method and Implementation We propose to learn Functional Simulation Affordance Fields (FSAFs), scene representations that embed rich semantic and visual features from a DINOv2 embedding model for arbitrary simulation objects and environments from OMNIGIBSON. Using our limited dataset of affordance labels (e.g. toggle buttons) in the simulation object prims, we then train a goal state regression model conditioned on this rich field model with embedded functional and affordance information. We then demonstrate the utility of our open-ended goal state annotations to collect trajectories from a motion planner and train a behavior cloning policy.

Results We learn FSAFs for 67 different object scenes in simulation environment, sampling 100,000 points from the embedding field. We qualitatively demonstrate the richness of all learned field models using both image and text queries.

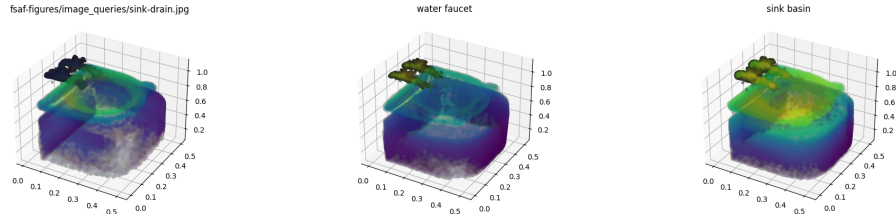


Figure 1: Image and text queries on a Functional Simulation Affordance Fields.

We then use our limited annotations to train a goal state regressor that achieves a validation MSE of 0.0395m.

Discussion Qualitative evaluation demonstrates the richness of the learned scene representation. Moreover, we quantitatively demonstrate that these representations can be leverage for downstream tasks, including goal states for producing trajectories for a behavior cloning policy on a single scene.

Conclusion We demonstrate the potential of FSAFs as scene representations, which can be leveraged to annotate relevant goal positions for task completion. We already observe success for robot learning and evaluation using this fairly limited annotation approach. Accordingly, we are optimistic about continuing work on this project to demonstrate the ability to validate robot behavior on arbitrary tasks, which in turn might shepherd the creation of measurably reliable, generalist, human-centric household robots.

Functional Simulation Affordance Fields

Aditesh Kumar

Department of Computer Science

Stanford University

aditesh@stanford.edu

Abstract

The burgeoning domain of Human-Centered AI is cautiously optimistic about the potential of the field of collaborative robotics. Simulation offers a dynamic benchmark environment in which open-ended human-sourced tasks can be concretely evaluated. However while existing simulations can create open-ended environments, they lack the detailed annotations necessary to properly measure robot behavior on arbitrary tasks in these environments. We propose learning Functional Simulation Affordance Fieldsscene representation that encode affordance and functional information for each point in a simulated environment. We demonstrate the effectiveness of these representations for learning arbitrary scene annotation tasks, which can then be used to train and evaluate robot policies on open-ended tasks.

1 Introduction

The current conversation around Human-Centered AI is cautiously optimistic about the potential of collaborative robotics. Spurred by recent advances in robotics and machine learning technology, robots are increasingly present in domestic and industrial spaces, raising concerns about how we can build autonomous systems that are "cooperative and well-aligned [with] humans' needs, goals, and values" Littman et al. (2022). While simulation offers a dynamic benchmark environment in which open-ended human-sourced tasks can drive the development and evaluation of human-centric collaborative robots, significant challenges remain in translating human-described tasks into concrete robot learning objectives.

The primary challenge lies in the gap between human task descriptions and actionable robot instructions. Simulations can serve as excellent environments for learning robot policies, but scaling learning to arbitrary tasks and scenes is difficult due to the necessity for manual inputs for goal states and rewards. However, the extent to which human-described tasks can be reconstructed for benchmarks faces severe limitations: simulation environments lack requisite annotations to provide concrete evaluations of robot performance on open-ended task definitions.

To address these limitations, we explore a means of learning semantic scene representations which can be leveraged for open-ended environment annotation and evaluation of generalist robot learning on arbitrary tasks. Specifically, we investigate two key research questions:

1. How can we learn to automatically annotate affordances in 3D objects in a way that facilitates robot learning in simulation?
2. How can these automatic labels be used to scale training of a policy to actuate a bimanual robot?

Our hypothesis is that we can leverage vision-language 3D scene representations to produce automated functional annotations for robot learning environments. Building on the robust foundation provided by the OMNIGIBSON simulator and the BEHAVIOR-1K benchmark environment Li et al. (2024), we learn Functional Semantic Affordance Fields (FSAFs) for diverse object scenes in simulation, sampling 100,000 points from each field. Through qualitative demonstration of the richness of all learned field models using both image and text queries, we show that our approach enables automated scene annotation. We then demonstrate the downstream viability of our annotations for scaling robot learning to arbitrary tasks and environments.

2 Related Work

2.1 Embodied AI Simulation and Benchmarks

In recent years, simulation has flourished as a venue for training and evaluating embodied AI policies for household environments. A simulation ecosystem for household robotics first proposed as IGIBSON in Shen et al. (2021) and Li et al. (2021), build on IsaacSim to create an ecosystem for training policies specifically to this domain. It was soon followed by Xia et al. (2020), which provided a necessary benchmark for evaluating policies learned in these new, simulated environments.

Notably, simulations have become especially promising evaluation environments for policies, with work like Shridhar et al. (2020) and Puig et al. (2018) also introducing different ways for measuring the behavior of embodied AI policies. A particularly notable effort along this line has been the BEHAVIOR benchmark Li et al. (2021), which takes a human-centric approach to and creates a benchmark of 1000 realistic, and diverse scenes and associated tasks, 900 of which were solicited from a wide-scale human survey. The work introduces the OMNIGIBSON simulator which provides primitives for robot learning, rich scenes and benchmarks, and complex physics simulation all in one environment. All items in the benchmark are hand-curated based on the human survey, and provide a narrow but diverse set of tasks for evaluation. The primary concern with this approach is the difficulty of scaling benchmark to novel scenes and tasks without extensive human input, especially for diverse, open-ended task formulations for which the simulator components lack sufficiently detailed annotations.

2.2 Scene Representations

Advances in scene representations and 3D reconstruction date back to the seminal work introducing Neural Radiance Fields Mildenhall et al. (2020). This approach synthesizes image features from multiple views to reconstruct a 3D model of a scene as a field, mapping coordinates to features. Language Embedding Radiance Fields Kerr et al. (2023) build upon this work and leverage vision-language embedding to introduce semantic information into the reconstructed representation. Specifically, LERF works by rendering CLIP embeddings across the different training image features for a radiance field, which makes a reconstructed field language-queryable. Since then, advances in vision-language embedding models have manifested more sophisticated representation learning in



Figure 2: Tiago robot functioning in simulated OMNIGIBSON environment.

the image domain. Notable work includes Oquab et al. (2024), Zhai et al. (2023), and Wu et al. (2025). DINOv2 in particular Oquab et al. (2024) scales self-supervised training to produce rich features that embed vision-language knowledge. While the original LERF work lacked some of the richness and specificity necessary to localize and bound exact features, these advanced models have demonstrated suprisingly effective downstream performance on a wide array of tasks due to their fine-grained feature learning approach.

3 Method

The following section outlines the three primary components of our approach: (1) Functional Simulation Affordance Fields, (2) goal state regressors, (3) and an imitation learning policy.

3.1 Functional Simulation Affordance Fields

The core of our approach, Functional Simulation Affordance Fields (FSAFs), are a field embedding model that employ multi-view synthesis of the scene. Specifically, we look to learn a function F that maps each coordinate in the simulation coordinate space (x, y, z) to an embedding of the affordances for that point in the scene.

$$F(x, y, z) \in \mathbb{R}^E$$

To capture this affordance embedding, we pan the simulation camera around the scene and sample RGB and depth images at different camera azimuths and elevations of the scene, cropping out patches at different resolutions. The RGB image is fed into a vision-language embedding model, in our case DINOv2, from which we extract final features for the various image patches. Using the depth image map as input, we train the model to embed semantic and visual information for each position in the scene using an MSE objective.

3.2 Goal State Regressor

To examine the viability of our approach, we sample an object/scene representation as embeddings from points on the surface on an object of interest. Concretely, this is a point cloud of embeddings:

$$z = F(\mathbb{R}^{H \times W \times D}) \in \mathbb{R}^{H \times W \times D \times E}$$

We use this representation to train a regression model that estimates the distance d_g to the nearest affordance for each of C classes of affordance present in a limited dataset of annotations we possess:

$$d_g = G(z_{i,j,k}) \in \mathbb{R}^C$$

We can then compute potential goal positions g_{pos} for each affordance as the local minima of the predicted distances over the entire field.

3.3 Training an Imitation Learning Policy

We collect motion planner trajectories for reaching g_{pos} from random starting positions and poses around an object that we have annotated with G . These trajectories are then used to train an imitation learning policy:

$$\pi(a|I_{rgb}, g_{pos})$$

The policy is trained to clone the motion planner using a simple MSE objective conditioned on the robot’s observations I_{rgb} .

4 Experimental Setup

OMNIGIBSON Li et al. (2024) provides detailed simulation environments built on top of IsaacSim. In this simulation environment, we sample 67 objects of interest. We learn FSAFs for these objects and then split the fields into a train and validation set. It takes around 45 minutes to learn each field using 2 NVIDIA TitanRTX GPUs (one for running the simulation and one for running the embedding models). During the learning process, we take 8 random crops from each camera view and stitch together 512 different views to learn the field embedding. We experiment with using DINOv2, SigLIP, and Sonata for the image embedding models, as well as with the amount of training time provided for each FSAF.

OMNIGIBSON provides limited annotation data for $C = 6$ affordance types for each object: {particlesink, togglebutton, openfillable, particlesource, fillable, heatsource}. Objects are often under-annotated, possessing 0-1 affordance positions for each class. We sample 100,000 points from each field model to produce z for each of the 67 objects. We then split these z into a training (80%) and validation set (20%), which we use to develop our goal position regressor model. We experimented with a naive class-agnostic affordance annotator and a class-based annotator, using different point cloud sampling and data splitting strategies.

For learning the policy, expert trajectories are collected using a cuRobo Sundaralingam et al. (2023) motion planner. We restrict the policy action space to the arms of the robot, randomizing the starting position and pose of the robot in the region around the object of interest. We use a simulated Tiago bimanual robot from PAL Robotics.

5 Results

We have experimented learning fields with multiple self-supervised text-image encoders, and initially saw relatively noisy results. Our refined field model was trained by randomly rotating a scene in the simulated environment, taking shots from random azimuths and elevations, cropped to look at different parts of the image. We then learned a field using the image features of these cropped images of the scene. Our best performing field model $F(x, y, z)$ used the image patch embeddings from the last layer of a DINOv2 encoder as features.

5.1 Quantitative Evaluation

Our best $G(z)$ model (Table 1) achieved an average 0.03 m error rate, which is well within the margin of error around which a simulated bimanual manipulator can operate. Key methodological discoveries that marked this approach were the division of the goal state regression objective by class. That is, producing different distance estimates for each kind of affordance, which we were able to demonstrably accomplish without losing performance by adding a layer to the network.

However, qualitative evaluation revealed flaws with our sampling strategy (Figure 3). These goal position estimators were trained over a volume of points sampled evenly over the entire scene, and learned to predict the L_2 distance to the nearest metalink. But some sampled points exist inside the volume of the perceived object; these points might be geometrically closer to the metalink, but are semantically quite far from them. Accordingly, it is wrong for our estimator to learn to predict that the inside of the oven of a range is somehow closer to the burner than a point on the edge of the top of the range. We resolved this by only sampling points from the field that lie on the surface of the perceived and training the objects to achieve our best model, after solving a minor data leakage problem that arose by splitting our train and validation sets by points as opposed to by object.

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Method	MSE (m)
Point-FC2-Volume-PointSplit	0.0796
MCPPoint-FC3-Volume-PointSplit	0.0587
MCPPoint-FC3-Surface-PointSplit	0.0001
MCPPoint-FC3-Surface-GridSplit	0.0395

Table 1: Validation error for goal position estimates.

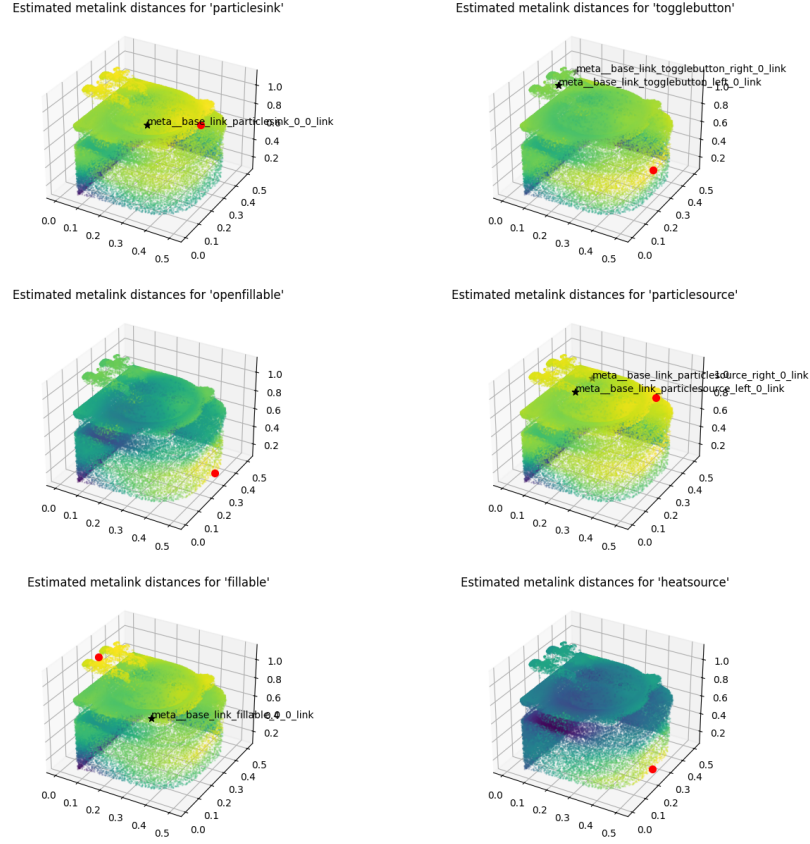


Figure 3: Estimates of distance to g_{pos} for six different metalink classes over a single learned field.

5.2 Qualitative Analysis

I also ran an experiment earlier on to see the effect of reducing the training time for the radiance fields on the quality of the generated fields (Figure 5). The key insight here was that reducing the number of training epochs from 100 to 75 might not yield a significant loss in quality to the field. This is beneficial because our dataset for training the goal distance estimator comes from learning fields over a diverse dataset of objects. For a dataset of different 67 objects, this procedure took about 8 hours on a cluster, using roughly 1 hour and two TitanRTX GPUs for each object. Cutting that time down 25% was a huge benefit to the project.

6 Discussion

Our results demonstrate that scene representations can effectively encode rich environmental features that serve dual purposes: enabling efficient querying for scene understanding and facilitating downstream task learning. The ability to train classifiers that produce reasonable estimates for arbitrary functional annotations represents a significant step toward flexible scene interpretation. Furthermore, our approach successfully leverages trajectories from deterministic planners to learn policies capable of reaching annotated goal states, establishing a viable pipeline from scene representation to task execution. This hierarchical framework shows promise for bridging the gap between scene understanding and actionable robotic behaviors.

However, our results raise more open questions that need to be addressed to fully examine the potential and limitations of Functio. The transferability of learned policies from simulation to real-world environments represents a fundamental challenge that will determine the practical applicability of our approach. Additionally, the scalability of our hierarchical method to long-horizon tasks remains unproven, as does our ability to effectively leverage the semantic knowledge embedded in scene representations for open-ended annotation tasks. Future work should investigate more efficient field model learning techniques and examine the failure modes of our imitation learning policies. The extent to which our goal state annotations make long-horizon household robotics tasks tractable for offline-RL methods also warrants further investigation.

7 Conclusion

The broad promise of simulation for real-world RL and household robotics is to scale evaluation of generalist policies. We demonstrate the potential of FSAFs as scene representations, which can be leveraged to annotate relevant goal positions for task completion. We already observe success for robot learning and evaluation using this fairly limited annotation approach. Accordingly, we are optimistic about continuing work on this project to demonstrate the ability to validate robot behavior on arbitrary tasks, which in turn might shepherd the creation of measurably reliable, generalist, human-centric household robots.

8 Team Contributions

This work builds off of ongoing work at the Stanford Vision and Learning Lab with my co-collaborators Auddithio Nag (aunag@stanford.edu) and Cem Gokmen (cgokmen@stanford.edu).

- **Aditesh Kumar:** In charge of designing the synthetic data generation for creating radiance fields, and scaling the training of these radiance fields to large datasets. I also handled the development of a metalink classifier to regress the positions of interactive features within the field, and for collecting trajectories from which I trained a behavior cloning policy. I also wrote this entire report.
- **Auddithio Nag:** Handled the technical experimentation for vision-language embedding models for conditioning the fields with information about affordances, and making them effectively queryable.
- **Cem Gokmen:** Leading the project, from coming up with the project idea to defining the methodological approach. Providing technical counsel on the integration of OMNIGIBSON.

I would also like to thank lab members Chengshu Li, Josiah Wong, and Hang Yin for their guidance on collecting policy trajectories in OMNIGIBSON.

Changes from Proposal In the proposal, I mistakenly framed FSAFs as a world modelling project. The technical approach has not changed as described then, but the terminology has been revised to correct this inaccuracy.

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A Additional Results

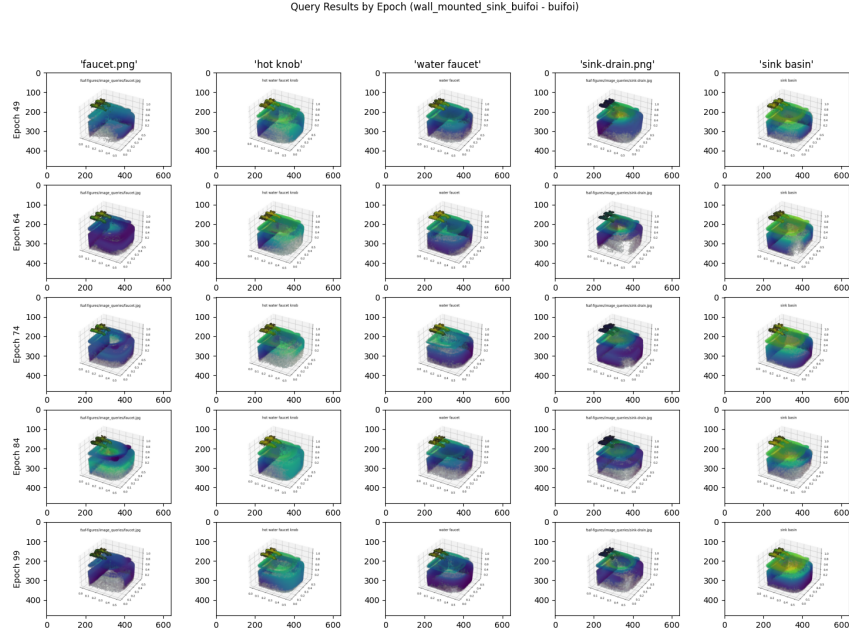


Figure 4: Query robustness for z at different epochs during training of the field model for a wall_mounted_sink, ID: buifoi.

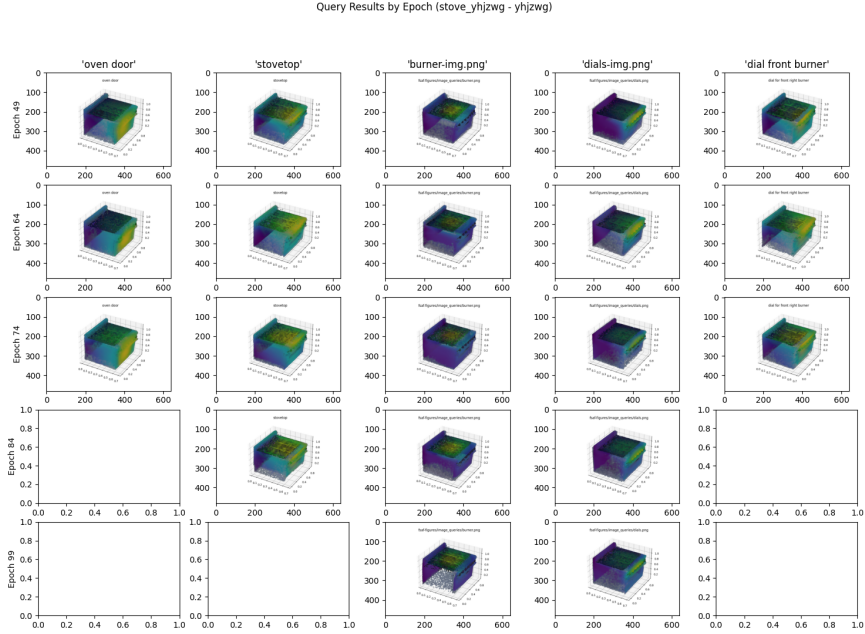


Figure 5: Query robustness for z at different epochs during training of the field model for a stove, ID: yhjpg.

B Implementation Details

The codebase for this project built off of the Python following packages.

Table 2: Python Package Requirements

Package	Version
numpy	1.26.4
omnigibson	1.1.1
torch	2.5.1
warp-lang	1.6.0
click	$\geq 8.0.0$
pydanclick	$\geq 0.5.1$
open3d	$\geq 0.19.0$
ipympl	$\geq 0.9.7$
polars	$\geq 1.23.0$
rerun-sdk	$\geq 0.22.1$
tqdm	$\geq 4.67.1$
transformers	$\geq 4.49.0$
lightning	$\geq 2.5.0$
tensorboard	$\geq 2.9.1$
sentencepiece	$\geq 0.2.0$
protobuf	$\geq 5.29.3$
robomimic	$\geq 0.3.0$
h5py	$\geq 3.10.0$